

# Lab 3 Deliverables: Component Modelling and Architecture Pattern Selection

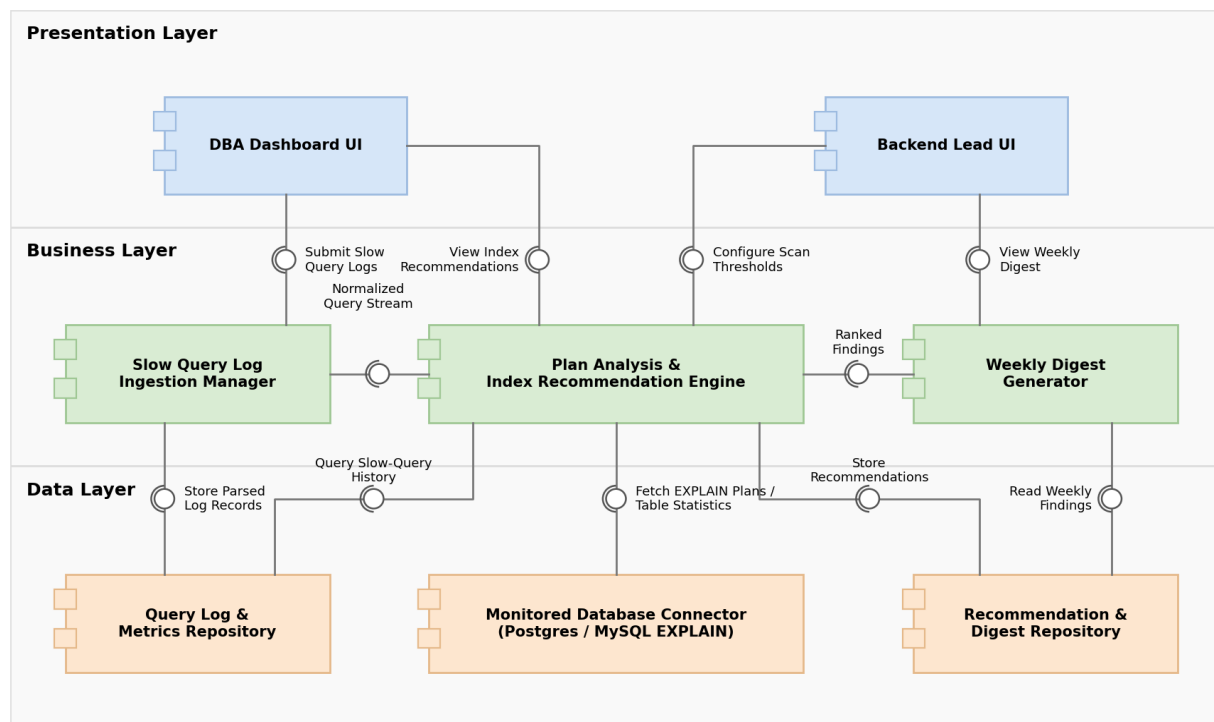
Name: Akshath Patil

SRN: PES1UG24CS044

Project Title: Database Query Performance Profiler

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## Architecture Selection and Diagram



## Justification

**Architecture Selection:** We chose Layered Architecture for the Database Query Performance Profiler system.

### Architectural Choice:

We selected the Layered Architecture pattern.

### Two Reasons:

**1. Clear Separation of Concerns:** This pattern isolates the user-facing interfaces (DBA Dashboard UI and Backend Lead UI) from the complex backend slow query log parsing, EXPLAIN plan analysis, index recommendation heuristics, and digest generation logic.

**2. Supports Traditional Workflows:** The step-by-step pipeline from slow query log submission (Presentation), to log parsing and normalization, sequential scan detection, missing index recommendation, and weekly digest generation (Business), to persistent storage of query metrics and recommendations (Data) maps directly onto horizontal layers.

### Security Advantage:

The architecture provides a significant security advantage by preventing direct database access from the Presentation layer. Because coupling is managed strictly between adjacent layers, the DBA Dashboard UI and Backend Lead UI cannot directly query or edit the Query Log & Metrics Repository or the Recommendation & Digest Repository, nor can they touch the credentials used to connect to the monitored Postgres/MySQL databases. All log submissions, scan thresholds, and recommendation requests must pass through the Business layer, ensuring that every action is validated and role-audited, and that sensitive data such as raw SQL text and connection credentials is masked and encrypted before reaching the Data layer.

### Performance Benefit:

A key performance benefit of this architecture is that the Business layer can act as a buffer. The Slow Query Log Ingestion Manager can stream, batch-parse, and normalize bursts of slow query log records (Postgres and MySQL formats) before passing them to the Query Log & Metrics Repository and the Plan Analysis & Index Recommendation Engine. Expensive EXPLAIN plan retrieval is done only by the Business layer through the Monitored Database Connector, in a controlled and rate-limited manner. This prevents the repositories from I/O saturation under peak log volumes and prevents the profiler from slowing the host database, ensuring the system handles up to 5,000 log records per minute without dropping entries (NFR-001).